INCH-POUND

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SUPERSEDING
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18 November 1980

MILITARY SPECIFICATION

RADIO SET AN/GRC-144( )(V)3 and (V)4

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of Radio Set AN/GRC-144(V)3 (HI-CAP LOS) and Radio Set AN/GRC-144(V)4 (SRWBR). It also applies to the AN/GRC-144(V)3 Radio Set as either totally manufactured end item or a modified AN/GRC-144(V)1 Radio Set.

1.2 Appendix. Appendix Radio Set AN/GRC-144(V)4 Short Range Wide Band Radio forms a part of this specification.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations,additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5000, by using the Standardization Document Improvement Proposal; (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5820

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SPECIFICATIONS

MILITARY

MIL-P-11268 - Parts, Materials and Processes Used in Electronic
MIL-M-13231 - Marking of Electronic Equipment Items
MIL-F-14072 - Finishes for Ground Electrical Equipment

STANDARDS

MILITARY

MIL-STD-252 - Wired Equipment, classification of Visual and Mechanical Defects
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781 - Reliability Tests; Exponential Distribution
MIL-STD-810 - Environmental Test Methods & Engineering Guidelines

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue Philadelphia, PA 1911-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

COMMUNICATIONS-ELECTRONICS COMMAND

SM-C-683640 Filter Low Pass, 4.5 MHz
SM-C-684447 Filter, Bandpass, 70 MHz
SM-C-684448 Filter, Low Pass, 1.0 MHz
SM-C-684449 Filter, Low Pass, 3.8 MHz
SM-D-956429 Attenuator Assy
SM-A-956742 Group A Electrical Performance Test Procedure for HI-CAP LOS/SRWBR Transmitter
SM-A-956744 Group B Performance Test Procedure for HI-CAP LOS/SRWBR Transmitter
SM-A-956746 Group C Electrical Performance Test Procedure for HI-CAP LOS/SRWBR Transmitter
SM-A-956748 Group A Electrical Performance Test Procedure for HI-CAP LOS/SRWBR Receiver
SM-A-956750 Group B Electrical Performance Test Procedure for HI-CAP LOS/SRWBR Receiver

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SM-A-956752	Group C Performance Test Procedure for HI-CAP LOS/SRWBR Receiver
SM-A-956754	HI-CAP LOS/SRWBR Radio Set Bounce Preconditioning Procedure
SM-A-956756	HI-CAP LOS/SRWBR Radio Set Burn-In Preconditioning Test Procedure
SM-A-956758	HI-CAP LOS/SRWBR Radio Set Operational Acceptance Test Procedure
DL-SM-B-956773	Radio Set, HI-CAP LOS AN/GRC-144(V)3
SM-A-966924	HI CAP LOS/SRWBR Radio Set Reliability Test Procedure

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3).

3.2. Construction. Radio set AN/GRC-144(V)3 shall be constructed in accordance with this specification and drawing and data list DL-SM-B-956773.

3.3. Parts, materials, and processes - general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4. Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, brackets, chassis, screws, etc., and necessary touch-up after mounting. All colored final film finishes shall match color chips provided by the procuring agency.

3.5. Marking (general). Marking shall conform to MIL-M-13231 and applicable construction drawings. Front panel marking shall be as specified in Table II, Group I as described in MIL-M-13231.

3.5.1. Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2. Identification numbers. All modules, sub-modules, assemblies, sub-assemblies, units and parts thereof shall be numbered where practicable to afford a means for configuration control and through-production traceability. The contractor, at his option, may prefix or suffix these numbers with a lot, batch or control number or letter according to his own requirements. These identification numbers may be rubber-stamped.

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3.5.3. Serial numbers. The following items shall have serial numbers:

Transmitter, Radio T-1054(P)A/GRC-144(V)
Receiver, Radio R-1467(P)A/GRC-144(V)
Digital Data Combiner Group OB-93/GRC-144(V)

3.6. Transmitter, Radio T-1054( )A/GRC-144(V).

3.6.1. Digital Data Combiner (DDC) characteristics.

3.6.1.1. Analog Voice Orderwire (AVOW) signal-to-noise (S/N) ratio. The signal-to-noise ratio for the Analog Voice Orderwire in a closed loop through the DDC shall be 25 dB minimum F1A weighted. In a closed loop, excluding the DDC, through the transmitter and receiver with rated deviation and a carrier-to-noise ratio of 13 dB, the S/N ratio shall be 32 dB minimum, F1A weighted (see 4.5.1).

3.6.1.2. Analog orderwire levels. The orderwire output levels shall be as listed in Table I for inputs to Recovered Orderwire, Local Handset Oscillator, Remote Phone In and Thru Orderwire (Receiver). In addition, the orderwire levels interfacing with the Orderwire Control Unit (OCU) shall be as follows:

a. Input level (Thru OW send). The full load input of 1 kHz test tone shall be +21.3 dBm ±1 dB.

b. Output level (Thru OW receiver). The full load output of 1 kHz test tone shall be -18.2 dBm ±1 dBm.

TABLE I. Analog OW levels.

1 kHz	Local Handset	Output (dBm) Remote Phone	XMTR OW	Thru OW
PCM Recovered OW -4 dBm @ 600 ohms ±10% balanced	-18 ± 3@ 150 ohms	-16 ± 3@ 600 ohms ± 10% balanced	-	-4 ± 3@ 600 ohms ± 10%
Local Handset Oscillator In 0 dBm @ 50 ohms	-18 ± 3@ 150 ohms	-16 ± 3@ 600 ohms ± 10% balanced	0 ± 3@ 600 ohms ± 10% balanced	-4 ± 3@ 600 ohms ± 10% balanced
Receiver In -11 dBm @ 50 ohms	-18 ± 3@ 150 ohms	-16 ± 3@ 600 ohms ± 10%	-	-4 ± 3@ 600 ohms ± 10% balanced
Remote Phone In -4 dBm @ 600 ohms ± 10% balanced	-18 ± 3@ 150 ohms	-	0 ± 3@ 600 ohms ± 10% balanced	-4 ± 3@ 600 ohms ± 10%
Thru OW (Receiver) In -4 dBm @ 600 ohms	-18 ± 3@ 150 ohms	-16 ± 3@ 600 ohms ± 10%	0 ± 3@ 600 ohms ± 10% balanced	

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3.6.1.3. Analog orderwire frequency response. The orderwire frequency response for the recovered and emergency orderwire levels shall be measured with inputs between 300 Hz to 1800 Hz at -4 dBm input level. The frequency response shall be:

Frequency (Hz)	Limit (dB)
300	-1.0, +2.0
500	-1.0, +1.8
700	-1.0, +1.8
900	+1.0
1000	(0) Reference
1200	+1.0
1400	+1.0
1600	+1.0
1800	+1.0

3.6.1.4. Analog orderwire peak limiting. The orderwire shall compress the input of the local and remote handset orderwire signal such that limiting will start at 0 dBm, the level required to produce rated orderwire deviation of the transmitter, with the following characteristics:

INPUT	COMPRESSION
10 db Below Initial Limiting	None
At Initial Limiting (0 dBm)	0.4 dB Max.
15 dB Above Initial Limiting	7 dB Minimum

3.6.1.5. Analog orderwire distortion. With a 0 dBm local handset input level, the distortion in any set of orderwire output terminals shall be less than 5% between 300 and 1800 Hz.

3.6.1.6. Video input impedance. The transmitter video input impedance shall be 50 ohms + 10%.

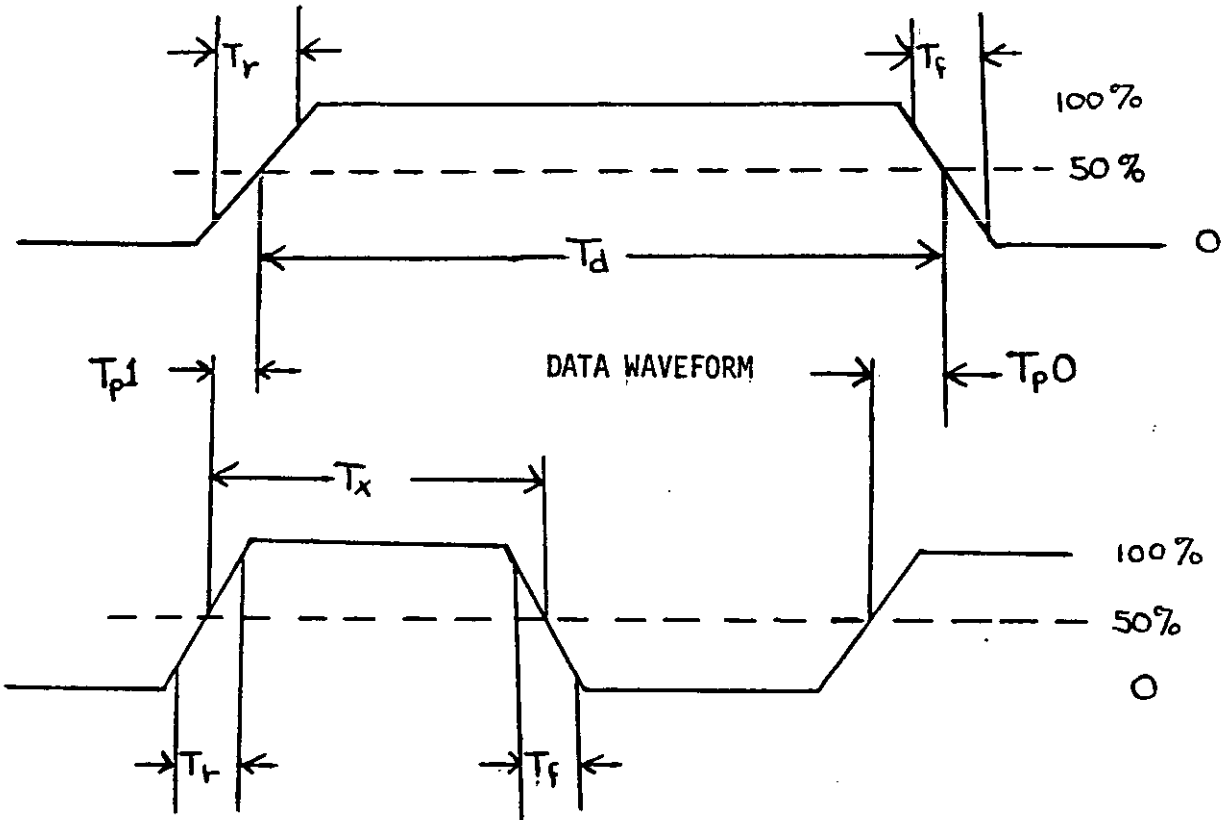
3.6.1.7. Analog Voice Order (AVOW) S/N ratio with supply line variations. The requirement of 3.5.1.1 shall be met for ac power conditions of 110 Vac, 47 Hz and 120 Vac, 63 Hz.

3.6.2. Digital Data characteristics.

3.6.2.1. Data rates. The data rates shall be 1.024, 1.152, 1.536, 2.048, 2.304, 4.096, 4.608 and 4.9152 mb/s. In addition, 9.36 and 18.72 Mb/s rates are provided for use in the SRWBR mode.

3.6.2.2. Format. The format shall be balanced NRZ data and associated square wave timing in accordance with figure 1.

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T_d = NOMINAL DATA BIT TIME = 1/BIT RATE
T_x = CLOCK PULSEWIDTH = $T_d/2$, NOMINAL. TOLERANCE PER TABLE II.
T_p1 = LOGIC 1 DELAY, PER TABLE II.
T_p0 = LOGIC 0 DELAY, PER TABLE II.
T_r = RISE TIME 10 - 90% PER TABLE II.
T_f = FALL TIME 10 - 90% PER TABLE II.

FIGURE 1. Wave forms for balanced digital signals

3.6.2.3. Driver output. The line-to-line minimum as measured at the end of 100 feet of RG-108 cable terminated in 100 ohms shall be +87 mV for a digital "1", -87 mV for a digital "0". Polarity as measured on the later lettered pin. The earlier letter pin is used as the reference. The RG-108 cable is used for test conditions only. The differential voltage levels in this paragraph are valid for operation of the driver into a load whose center-tap is either grounded or ungrounded. The maximum voltage measured from either line to ground shall not exceed ±3 volts. Line-to-line voltage maximum measured directly into a 100 ohm termination shall not exceed ±4 volts.

3.6.2.4. Rise and fall times. Rise and fall times measured in a 100 ohm resistive dummy load at the driver output terminals shall be in accordance with figure 1 and table II.

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TABLE II. Rise and fall times.

Maximum Bit Rate	Delays (ns max) Tp	Rise or Fall Time (ns max) Tf or Tr	Clockwidth Tolerance (ns max) Tx
2 kb/s	±1500	3000	±12,000
16 kb/s	±1500	3000	±1500
4.9152 Mb/s	±29	30	±15
18.72 Mb/s	±8	8	±4

3.6.2.5. Clock data phasing. Clock data phasing shall be in accordance with figure 1 and table II.

3.6.2.6. Driver offset. The magnitude of the generator offset voltage, Vos, measured between the center point of the test load and generator circuit ground (terminal C) shall not be greater than 3 volts. (See figure 2).

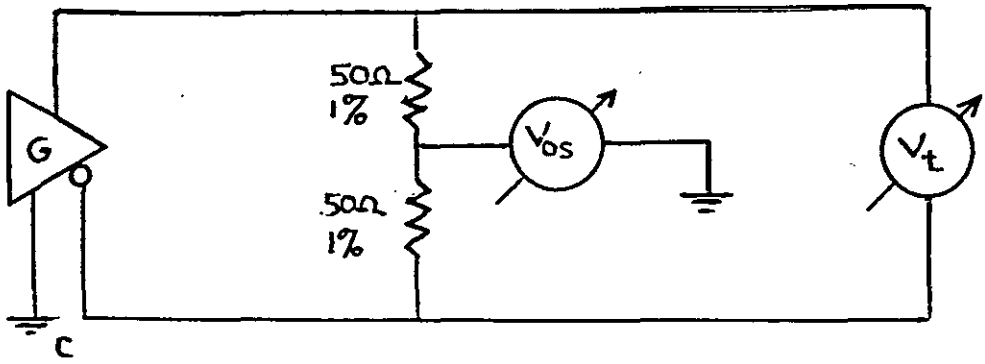


FIGURE 2. Measurement of generator offset

3.6.2.7. Signal balance. When the binary state changes, the polarity of Vt (see figure 2) shall be reversed (Vt). The magnitude of Vt shall differ by less than 10% of Vt.

3.6.2.8. Ringing. When 100 ft. of RG-108 A/U shielded terminated twisted pair, or equivalent, is connected to the driver output, the waveform at the line output shall not overshoot its final value by more than 15%. Undershoot should not reduce the signal to less than 87 mV.

3.6.2.9. Receiver sensitivity. The receiver shall deliver the proper digital output when the line-to-line voltage of +50 mV for a digital "1" or 50 mV for a digital "0" is applied at the receiver input terminals.

3.6.2.10. Input load impedance. 100 ohms +10% line-to-line.

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3.6.2.11. Digital Voice Orderwire (DVOW). The DDC shall provide a full-duplex serial DVOW signal (at the OCU interface connector) as follows:

3.6.2.11.1. Rate. The rate shall be 16 kb/s.

3.6.2.11.2. Format. The format shall be balanced NRZ data and associated square-wave timing signal in accordance with figure 1 and 3.5.1.8.3 through 3.5.1.8.10.

3.6.2.12. Data Orderwire (DOW). The DDC shall provide a full duplex serial digital orderwire signal (at the OCU interface connector) as follows:

3.6.2.12.1. Rate. The rate shall be 2 kb/s.

3.6.2.12.2. Format. The format shall be balanced NRZ data and associated square-wave timing signal in accordance with figure 1 and 3.5.1.8.3 through 3.5.1.8.10.

3.6.3. Alarm systems. The following alarm systems shall be provided.

3.6.3.1. Transmitter panel mounted alarms. These alarms shall provide local visual and audible, and remote audible and closure alarm indications for transmitter fault conditions as shown in Table III (see 4.5.1).

TABLE III. Transmitter alarms.

Alarm Indicator	Function	Indicator Color	Test Condition
Radio Test Set	Status of Radio Test Set	GREEN	Radio Test Set Off
Wave Guide Switch	Status of Waveguide Switch	RED	Radio Test Set On
		GREEN	Transmitter Output to Antenna
		RED	Transmitter Output to Load
Ring	Status of Signalling	GREEN	No Signalling
		AMBER	Signalling (Remote or Local)
Ring (Switch) Depressed)	Status of Signalling	AMBER	Speaker emits tone signal and remote phones ring
Central Alarm	Status of all Equipment (XMTR, RCVR, & External Traffic)	GREEN	All equipment normal
		RED	Failure of one or more equipment

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TABLE III. Transmitter alarms. - Continued

Alarm Indicator	Function	Indicator Color	Test Condition
Traffic	Status of all Equipment FM Deviation	GREEN	Deviation normal
		RED	Deviation below that required or no traffic
RF Power	Status of Transmitter Power Output	GREEN	Output power normal
		RED	Output power failure
Refl RF Power	Status of Transmitter Reflected RF Power	GREEN	Refl power normal
		RED	Refl pwr excessive
Synthesizer Lock	Status of Synthesizer Phase Lock	GREEN	Normal Condition
		RED	Out of Lock
Radio-to-Cable Modem	Status of Modem in One Direction	GREEN	Normal Operation
		RED	Failure functional
Cable-to-Radio Modem	Status of Modem in Other Direction	GREEN	Normal operation
		RED	Failure functional

3.6.3.2. Shelter mounted alarms. The following shelter mounted traffic alarms (not part of the Radio Set) are powered (+28 Vdc) and operated by the transmitter (See Table IV).

TABLE IV. Shelter mounted alarms.

Alarm Indicator	Function	Indicator Color	Test Condition
Traffic in	Status of traffic coming in	GREEN	Normal traffic
		RED	No traffic in
Traffic out	Status of traffic going out	GREEN	Normal traffic
		RED	No traffic out

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3.6.3.3. Orderwire Control Unit (OCU) alarms. The following alarms (closures) are sent to the OCU.

DDC Summary alarm (SA)

DDC Loss of received radio signal (RS)

Receiver Summary alarm

3.6.4. Front panel meter functions. The meter functions shall be in the indicated color band with the Refl RF Power meter indicating less than 30. The "Off-Transit" position shall indicate zero.

3.6.5. Deviation capability. The deviation shall be adjustable such that the deviation (Traffic) meter shall indicate 100 + 10 divisions and the rated peak deviations shall be as follows:

Data Rate (Mb/s)	Peak Deviation (MHz)
1.024 to 2.304	+ 0.72
4.096 to 4.9152	+ 1.75

3.6.6. Power output. The transmitter power output shall be greater than 0.25 watts (250 mW) through the frequency range of 4.4 to 5.0 GHz when operated into a matched load. The power output shall be no less than 0.16 watts (160 mW) during or 0.20 watts (200 mW) after any environmental test.

3.6.6.1. Power output with supply line variations. With ac power source adjusted between 110 Vac, 47 Hz and then to 120 Vac, 63 Hz, the requirement of 3.5.5 for transmitter power output shall be met.

3.6.7. Frequency accuracy. The carrier frequency accuracy shall be + 40 kHz of the desired frequency.

3.6.8. Radio test set operation. The radio test set output shall be -33 dBm minimum at 4.7 GHz with its controls set for HI output level, and -63 dBm minimum at LOW output level.

3.7. Receiver, Radio R-1467( )A/GRC-144(V).

3.7.1. Noise figure. The receiver noise figure throughout the tunable frequency range shall be 5.5 dB plus 0.3 dB, minus any.

3.7.2. Output capability and stability. The baseband output capability of the receiver operating into a 50 ohm termination, shall be adjustable to 1.0 volt peak to peak. The output level shall not vary by more than + 1 dB with input levels between -88.5 dBm to -38.5 dBm at room ambient and +2 dBm during or after any environmental test.

3.7.3. Alarm systems. The alarms shall provide local visual and audible and remote closure and audible alarm indications for receive fault conditions. The following alarm systems shall be provided (see table V):

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TABLE V. Receiver alarms.

Alarm Indicator	Function	Indicator Color	Test Condition
Carrier (IF)	Status of carrier signal	GREEN	Normal carrier signal
		RED	Carrier signal removed
Synthesizer Lock	Status of synth Lock	GREEN	Normal synth lock condition
		RED	Synth unlocked
Traffic	Status of traffic output	GREEN	Normal traffic output signal
		RED	Loss of traffic output signal

3.7.4. Meter functions. The meter functions shall be in the indicated color band or as otherwise specified. The "Off-Transit" position shall indicate zero.

3.7.5. Bandwidth. The 3 dB IF receiver bandwidth shall be 5.0 MHz + 500 kHz for traffic rates from 1.024 to 4.9152 Mb/s.

3.7.6. Carrier level threshold. At 4.9152 Mb/s the receiver threshold shall be reached by signals no greater than -88.5 dBm.

3.7.6.1. Carrier level threshold with supply line variations. With the ac power source adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz, the requirement of 3.6.6 shall be met.

3.7.7. Frequency accuracy. The maximum error of the local oscillator frequency measured at the output of the local oscillator filter (FL-6) shall be + 15 kHz.

3.7.7.1. Frequency accuracy with supply line variations. With the ac power source adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz, the requirement of 3.6.7 shall be met.

3.8. Transmitter/Receiver.

3.8.1. Operating range. Minimum range of the AN/GRC-144(V)3 in the 4.4 to 5.0 GHz band shall be 40 km for data rates up to 4.9152 Mb/s.

3.8.2. Channel spacing. The AN/GRC-144(V)3 shall have 6001 channels, each spaced 100 kHz apart, in the transmit and receive directions.

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3.8.3. System bandwidth. The overall transmitter through receiver bandwidth shall be optimized in two bandwidths for transmission of data up to 4.9152 Mb/s: (see table IV).

Data rate (Mb/s)	Bandwidth (MHz) 3 dB + 0.6 dB
1.024 to 2.304	0.92
4.096 to 4.9152	1.90

TABLE VI. Mode module implementation.

Module Location				
Radio Usage	Transmitter Unit 1		Receiver Unit 2	
	Refer des A3	Ref des A6	Ref des A3	Ref des A6
1-2 Mb/s	SM-D-956429 Attenuator Assy	SM-C-683640 Fltr, LP, 4.5 MHz	SM-C-684448 Fltr, LP, 1.0 MHz	SM-D-684447 Fltr, BP 70 MHz (B= 5.0 MHz)
4-5 Mb/s	SM-D-956429 Attenuator Assy	SM-C-683640 Fltr, LP, 4.5 MHz	SM-C-684449 Fltr, LP, 3.8 MHz	SM-D-684447 Fltr, BP 70 MHz (B= 5.0 MHz)

3.8.4. Bit error rate. The transmitter and receiver operating in a closed loop through the digital combiner shall have bit errors of less than 1 in 105 for an RF simulated path loss of 107.7 dB + 0, -1 dB in the 4.4 to 5.0 GHz band at the following rates:

3.8.4.1. Traffic. 1.024, 1.152, 1.536, 2.048, 2.304, 4.096, 4.608 and 4.9152 Mb/s.

3.8.4.2. DVOW. 16 kb/s for the traffic data rates in 3.7.4.1.

3.8.4.3. DOW. 2 kb/s for the traffic data rates in 3.7.4.1.

3.8.4.4. Data characteristics. The data characteristics shall be in accordance with 3.5.1.8.

3.8.5. Transmitter and receiver system PCM phase response. The system phase response shall be selectable to have an absolute phase shift of 0 degrees or 37 + 10 degrees at 300 Hz. The amplitude response at 300 Hz shall be 1 dB maximum below the 1 kHz level.

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3.8.6. Video frequency response. The video frequency response, with the low pass filters as listed in Table VI, and using a test transmitter operating at rated deviation, shall be as listed in table VII.

TABLE VII. Video frequency response.

Frequency (kHz)		
1-2 Mb/s	4-5 Mb/s	Relative response (dB)
50	50	0 (Reference)
460	950	1.0 dB max.
650	1330	1.5 + 0.5 dB
920	1990	3.0 + 0.6 dB
1300	2660	6.0 + 0.9 dB
1840	3800	12.0 + 1.2 dB

3.8.7. Signal-to-noise ratio. The receiver output of the radio set shall have a minimum of 20 dB signal-to-noise (peak signal to RMS noise) ration performance for digital data operation from 1 to 5 Mb/s. The RF simulated path loss shall be 107.7 + 0, -1.0 db in the 4.4 to 5.0 GHz band.

3.8.8. DDC operation with supply line variations. With the ac power source adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz the requirements of 3.7.4 shall be met.

3.8.9. Operational test (Radio Set). The operational test of 4.14 shall be performed to determine that the radio set is operating satisfactorily before, during and after environmental tests and to determine that each radio set is operating satisfactorily prior to packaging.

3.9. Preconditioning.

3.9.1. Bounce preconditioning. All new equipments or new and modified radio modules shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7.1.

3.9.2. Electrical preconditioning (Burn-in). All new equipments or new and modified radio modules shall be subjected to a forty eight (48) hour failure-free burn-in test. Normal or simulated operating signals shall be applied to the equipment during the operational preconditioning and periodic checks performed to detect any failures that may occur. During this preconditioning, the units shall be periodically de-energized and re-energized for a minimum of six (6) times. (Off-time: 2 minutes; On-time: 2 hours minimum) (See 4.7.2).

3.10. Reliability. The AN/GRC-144(V)3 shall have a specified (θ°) Mean-Time Between-Failure (MTBF) equal to or greater than 600 hour when tested under the conditions as listed in 4.11 herein.

3.11. Service conditions. The equipment shall meet the following service conditions. Where a test is referenced, meeting the test shall be considered

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as compliance with the requirement. The equipment shall meet full specification performance of the measurement specified.

3.11.1. Temperature.

3.11.1.1. Low (See 4.10.1.1).

a. Operating. The equipment shall be operable without degradation in specified performance at ambient temperatures as low as -25°F (-31.7°C).

b. Storage and transportation. The equipment shall withstand exposure to air temperatures as low as -65°F (-53.8°C).

3.11.1.2. High (See 4.10.1.2).

a. Operating. The equipment shall be operable without degradation in specified performance at ambient air temperature as high as +125°F (51.6°C).

b. Storage and transportation. The equipment shall withstand exposure to ambient air temperatures as high as +160°F (71.1°C).

3.11.2. Humidity. The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas. (See 4.10.2).

3.11.3. Altitude. The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level and shall withstand air transportation at altitudes up to 40,000 feet above sea level. (See 4.10.3).

3.11.4. Bench handling. The equipment shall withstand shock encountered in servicing. (See 4.10.4).

3.11.5. Bounce, vehicular. The equipment shall be mounted in its transport packing and shall be capable of withstanding 10g shock input of a random nature for a period of three (3) hours. (See 4.10.5).

3.11.6. Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.10.6. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface. (See 4.10.6).

3.12. Electromagnetic interference characteristics. The complete equipment AN/GRC-144(V)3 including the DDC shall meet the following EMI and susceptibility requirements of MIL-STD-461.

Test Method	Description	Frequency Range, Hz	Notes
CE01	Primary Powerline Emissions	30 to 20k	(1),(2),(3)
CE02	Control and Signal Line Emissions	30 to 20k	
CE03	Primary Powerline Emissions	20k to 50M	

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CE04	Control and Signal Line Emissions	20k to 50M	
CE06	Spurs at the Antenna Terminal	3.5G - 12.4G	(4)
CS01	Low Frequency Susceptibility, Primary Power Lines	30 to 50k	
CS02	High Frequency Susceptibility, Primary Power Lines	50k to 400M	
CS03	Intermodulation Susceptibility in RF Receivers	3.5G - 12.4G	
CS04	Input Selectivity Susceptibility in RF Receivers	3.5G - 12.4G	
CS05	Cross-Modulation Susceptibility in RF Receivers	3.5G - 12.4G	
CS06	Spike Susceptibility in Primary Power Lines	10 μs, + 110V	
RE02	E-Field Emissions from Case and Cabling	14k to 10G	(5)
RS03	Case and Cable Susceptibility to External E-Fields	14k to 10G	

NOTES: (1) CE01 Broadband emission limit for RCVR Cabinet: +97 dBua/20 kHz, based on a 2.3-ampere primary current.

(2) CE01 Broadband emission limit for XMTR Cabinet: +98 dBμ/20 kHz, based on a 3.0-ampere primary current.

(3) Additional relief, CE01 Broadband emission limit: Additional relief shall be afforded the limits of (1) and (2) above, in accordance with the following formula:

$$\text{(Delta relief, dB} = 20 \log) 10 \frac{\text{(Primary current of SRWBR Cabinet)}}{\text{(Applicable current of (1) or (2), above)}}$$

(4) Test Method CE06: Reference MIL-STD-461. Paragraphs 6.3.1 and 6.3.2 are not applicable to this test. Paragraph 6.3.3 is applicable to this test.

(5) RE02, Narrowband: The following relief factors are in effect for this test method:

(a)	14 kHz to 1 GHz	6 dB
(b)	1 GHz to 5 GHz	20 dB for all emissions except the C-Band transmit signal, and the 4th harmonic of the Transmitter's frequency synthesizer.

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(c) C-Band Transmit Signal: 30 dB

(d) 4th Harmonic of the Transmitter's Frequency Synthesizer: 25 dB

3.13. Systems safety engineering.

3.13.1. Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection. (See 4.6).

3.13.2. Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.13.3. Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.14. Interchangeability. Like units, modules, chassis, assemblies, sub-assemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454. (See 4.13).

3.15. Workmanship. The equipment shall be manufactured and assembled in accordance with requirements 5 and 9 of MIL-STD-454 and the applicable portions of the following paragraphs of MIL-P-11268. (See 4.6.).

General requirements for plastic material and plastic parts wiring and cabling, including:

Slack
Protection
Clearance
Splicing
Soldering and Brazing
General requirements for securing of parts
Cleaning of units
Process for welding
Controls
Self-tapping screws
Cleaning prior to welding
Connections, general
Grounding, general
Shielding on wire and cable

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the

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contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4.).

c. Quality conformance inspection of equipment before packaging. (See 4.5.).

4.3 First article inspection. This inspection shall consist of all the inspection specified in Table VIII and shall be performed in the order shown on the number of sample units specified herein, unless otherwise specified in the contract.

TABLE VIII. Order of first article inspection.

INSPECTION	REQ. PARA	INSP. PARA	UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5-8	Notes
1. Inspection covered by subsidiary documents.		4.4	Inspections to be performed on all units.					c
2. Bounce Pre-conditioning	3.9.1	4.7.1	Inspections to be performed on all units.					c
3. Electrical Preconditioning	3.9.2	4.7.2	Inspections to be performed on all units.					c
4. Group A Inspection	See Table IX		Inspections to be performed on all units.					c

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TABLE VIII. Order of first article inspection - Continued.

INSPECTION	REQ. PARA	INSP. PARA	UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5-8	Notes
5.Group B Inspection	See Table X		1	1				a
6. Group C Inspection (Electrical)	See Table XI		1	1				a
7. EMI	3.12	4.12	2					-
8. Service Conditions								
Non-damaging:								
Low Temp.	3.11.1.1	4.10.1.1		2				-
High Temp.	3.11.1.2	4.10.1.2		3				-
Altitude	3.11.3	4.10.3			1			d
Bench Handling	3.11.4	4.10.4			2			c
Potentially damaging:								
Humidity	3.11.2	4.10.2		4				c
Bounce Veh	3.10.5	4.10.5			3			d
Fungus	3.11.6	4.10.6				1		d
9. Reliability	3.10	4.11					1	b

NOTES:

a. The inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting them to any other inspection requirements specified in the table except that Group B and Group C inspections may be commingled with Group A inspections to reduce the number of test set-ups. Unless otherwise noted the unit is an AN/GRC-144 modified for HI-CAP LOS SRWBR operation or a new AN/GRC-144(V)3.

b. Reliability qualification (demonstration) testing for the AN/GRC-144(V)3 shall be conducted on first article units (4 units shall be the sample size required for qualification testing) in accordance with 4.11.

c. New radio set or new and modified modules only.

d. New radio set, no test required for modified AN/GRC-144.

4.4. Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before packaging:

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ITEM	WHERE REQUIRED
Finish	3.4
Marking	3.5
Safety	3.13
Workmanship	3.15

4.5. Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1. Group A inspection. Each radio set contract shall be inspected for conformance to the inspections specified in the Table IX. The order of inspection of the subassemblies, assemblies and units shall be as follows:

a. Prior to their assembly into the end equipment, the radio set subassemblies on order (new and modified subassemblies only) shall be subjected to a 100% visual and mechanical inspection by the contractor for conformance with the inspection requirements provided in Table IX.

b. Assemblies from the discrete lots which have passed the above inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 3.9 and a 100% visual/mechancial and electrical inspection in accordance with Table IX.

c. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection in accordance with Tables IX-1 and IX-2. Group A electrical inspection shall be performed in any order which is satisfactory to the Government, except that the operational test 4.14 shall be performed last (See 6.7).

NOTE: Assemblies are not required to undergo further visual/mechanical inspection as part of the contractor's 100% Group A inspection of the end equipment.

TABLE IX. Group A inspection.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
Visual/mechanical:		
Cabinets, elec. equipment	3.13 3.15	4.6
Printed wiring assemblies	3.13 3.15	4.6
Electrical:		
Analog Voice Orderwire (AVOW) signal-to-noise	3.6.1	SM-A-956742

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TABLE IX. Group A inspection - Continued.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
(S/N) ratio Transmitter panel	3.6.3.1	SM-A-956742
Shelter mounted alarms	3.6.3.2	SM-A-956742
Orderwire Control Unit (OCU) alarms	3.6.3.3	SM-A-956742
Front panel meter functions	3.6.4	SM-A-956742
Deviation capability	3.6.5	SM-A-956742
Power output	3.6.6	SM-A-956742
Noise figure	3.7.1	SM-A-956748
Alarm systems	3.7.3	SM-A-956748
Meter functions	3.7.4	SM-A-956748
Carrier level threshold	3.7.6	SM-A-956748
Bit error rate	3.8.4	SM-A-956742
Operational test (Radio Set)	3.8.9	SM-A-956758

Note a: Defects in all characteristics other than visual/mechanical are considered major.

TABLE IX-1. Sampling plan for visual and mechanical inspections.

Lot Size	Sample Size
2 to 8	*
9 to 15	8
16 to 25	8
26 to 50	8
51 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27
1201 to 3200	35
3201 to 10,000	38

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TABLE IX-1. Sampling plan for visual and mechanical inspections - Continued.

Lot Size	Sample Size
10,001 to 35,000	46
35,001 to 150,000	56
150,001 and over	64

NOTES:
a. * Indicates entire lot must be inspected
b. Acceptance number in all cases is zero.

TABLE IX-2. Sampling plan for electrical inspections.

Lot Size	Sample Size
2 to 8	5
9 to 15	5
16 to 25	5
26 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40

NOTE:
a. Acceptance number in all cases is zero.

4.5.2. Group B Inspection. Group B inspection shall be performed on lots that passed Group A inspection. This inspection shall consist of the inspections listed in Table X. Each lot shall be subjected to sampling inspection in accordance with Table X-1. Lots in which samples exhibit any failures shall be screened for that failure prior to the units within the lot being subjected to Group C inspection.

TABLE X. Group B inspection.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
Interchangeability	3.14	4.13
Analog orderwire levels	3.6.1.2	SM-A-956744
Bandwidth	3.7.5	SM-A-956750

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TABLE X-1. Sampling plan for group B inspections.

Lot Size	Sample Size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29

NOTE:
a. Acceptance number in all cases is zero.

4.5.3. Group C Inspection. Group C inspection shall be performed on units that have passed Group A and B inspection. The inspection shall consist of the inspections specified in Table XI.

4.5.3.1. Sampling for inspection.

4.5.3.1.1. Subgroup I. For this subgroup, two (2) units from the first month's production shall be selected for the first Group C inspection. For subsequent Group C inspections, two (2) from each successive 100, or two (2) units per month if less than 100 are produced monthly, shall be selected. For this subgroup, the maximum number of units to be selected and tested in a single month shall not exceed eight (8) units.

4.5.3.1.2. Subgroup II. For this subgroup, two (2) units from the first month's production shall be selected for the first Group C inspection. For subsequent Group C inspections, two (2) units shall be selected and tested during each two (2) month period throughout production.

4.5.3.1.3. Subgroup III. For this subgroup, excluding First Article testing, one (1) unit shall be selected at random, and tested at midpoint during the production schedule.

TABLE XI. Group C inspection.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
Subgroup I Analog orderwire frequency response	3.6.1.3	SM-A-956746

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TABLE XI. Group C inspection - Continued.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
Analog orderwire peak limiting	3.6.1.4	SM-A-956746
Analog orderwire distortion	3.6.1.5	SM-A-956746
Video output impedance	3.6.1.6	SM-A-956746
Analog Voice Orderwire (AVOW) S/N ratio with supply line variations	3.6.1.7	SM-A-956746
Digital data characteristics:	3.6.2	SM-A-956758
Data rates	3.6.2.1	"
Format	3.6.2.2	"
Driver output	3.6.2.3	"
Rise and fall times	3.6.2.4	"
Clock data phasing	3.6.2.5	"
Driver offset	3.6.2.6	"
Signal balance	3.6.2.7	"
Ringing	3.6.2.8	"
Receiver sensitivity	3.6.2.9	"
Input load impedance	3.6.2.10	"
Digital Voice Orderwire (DVOW)		SM-A-956758
Rate	3.6.2.11.1	"
Format	3.6.2.11.2	"
Data Orderwire (DOW)	3.6.2.12	"
Rate	3.6.2.12.1	"
Format	3.6.2.12.2	"
Power output with supply line variations	3.6.6.1	SM-A-956746
Frequency accuracy	3.6.7	SM-A-956746
Radio test set operation	3.6.8	SM-A-956746
Output capability and stability	3.7.2	SM-A-956752
Carrier level threshold line variations	3.7.6.1	See Note a
Frequency accuracy	3.7.7	SM-A-956752

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TABLE XI. Group C inspection - Continued.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
Frequency accuracy with supply line variations	3.7.7.1	See Note b
Operating range	3.8.1	SM-A-956752
Channel spacing	3.8.2	SM-A-956752
System bandwidth	3.8.3	SM-A-956752
Transmitter and receiver system PCM phase response	3.8.5	SM-A-956752
Video frequency response	3.8.6	SM-A-956752
Signal-to-noise ratio	3.8.7	SM-A-956752
DDC operation with supply line variations	3.8.8	SM-A-956752
Subgroup 2 (See Note c and d)		
Temperature	3.11.1	
High	3.11.1.2	4.10.1.2
Low	3.11.1.1	4.10.1.1
Humidity	3.11.2	4.10.2
Bench handling	3.11.4	4.10.4
Bounce, vehicular	3.11.5	4.10.5
Electromagnetic interference	3.12	4.12
Subgroup 3		
Altitude (See Note d)	3.11.3	4.10.3
Fungus	3.11.6	4.10.6
Electromagnetic	3.12	4.12

Notes:

a. Test to be performed in accordance with SM-A-956748 except that the ac power source shall be adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz.

b. Test to be performed in accordance with SM-A-956752 except that the ac power source shall be adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz.

c. Both units shall be inspected for the following EMI characteristics:

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Receiver Mode: CE06, RE02 and RS03
Transmit Mode: RE02

After EMI testing one unit shall be subjected to the high temperature, low temperature and humidity tests in that order. The remaining unit shall be subjected to the bench handling and bounce, vehicular tests in that order.

d. Perform the operational test before, during (as applicable) and after environmental test in accordance with the applicable test procedure.

4.5.3.2. Group C failures. Actions required relative to Group C failures **all be as specified in the contract.

4.5.4. Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5. Group D inspection. This inspection shall consist of the tests specified in Table XII and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE XII. Group D inspection.

INSPECTION	REQUIRED PARAGRAPH	TEST PROCEDURE
Reliability	3.10	SM-A-966924

4.5.5.1. Sampling for Group D inspection. Four (4) samples shall be selected at random from the monthly production lot.

4.5.5.2. Group D failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.5.3. Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on contract provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.6. Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.13 which can be determined visually. (See 3.13 and 3.15).

4.7. Preconditioning. (See 3.9).

4.7.1. Bounce preconditioning. All new radio sets (or new and modified modules only) shall be subjected to the following test:

a. The radio set or new and modified radio set modules shall be placed in their normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or

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equal. The package tester, shafts in phase, shall have a speed such that it is possible to insert a 1/32 - inch-thick strip of material under one corner or edge of the unit to distance of 3 inches as the unit bounces. The equipment shall be subjected to this preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspections. (See also SMA-A-956754).

4.7.2. Electrical preconditioning. All radio sets, or new and modified radio set modules shall be subjected to a forty-eight (48) hour failure-free burn-in test. Normal or simulated operating signals shall be applied to the equipment during the operational pre-conditioning and periodic checks performed to detect any failures that may occur. During this pre-conditioning, the units shall be periodically de-energized for a minimum number of six (6) times. (off-time: 2 minutes; on-time: 2 hours minimum.) (See also SM-A-956756).

a. All radio sets, or new and modified radio set modules on test shall be considered to have been successfully electrically pre-conditioned when they have individually passed a cumulative 48-hour operating period without failure. If during the 48-hour-operating pre-conditioning period a failure occurs, repair may be made using pre-conditioned sub-assemblies or modules and the pre-conditioning continued without the restart of a new 48 pre-conditioning period. A sub-assembly or module failing operating pre-conditioning shall, after repair, be resubmitted to and pass the entire 48-hour-operating pre-conditioning period prior to final electrical assembly tests by the contractor.

b. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical pre-conditioning, may be repaired with an electrically preconditioned part and electrically retested, accepted and delivered without repeat of electrical pre-conditioning.

4.8. Standard test conditions. Unless otherwise specified, all measurements and tests shall be conducted as follows:

4.8.1. Standard ambient conditions. The standard ambient conditions are:

Temperature	23° ± 10°C (73° ± 18°F)
Relative Humidity	Uncontrolled
Atmospheric Pressure	725 + 50 -115 mm. of mercury (28.5 + 2.0 - 4.5 in. of mercury)

4.8.2. Standard line voltage and frequency. Unless otherwise specified, all measurements and tests shall be conducted with a line voltage of 115 ± 5 VAC, single phase, 60 Hz.

4.9. Electrical tests. Electrical tests shall be performed in accordance with the applicable test procedures of the appropriate table.

4.10. Service condition tests. The operational test shall be performed during the periods specified in the applicable test procedures.

4.10.1. Temperature.

4.10.1.1. Low. The equipment shall be tested in accordance with Method 502.2, Procedure I of MIL-STD-810. The storage temperature shall be -65°F,

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(-53.8°C) and the operating temperature shall be -25°F (-31.7°C). The storage time shall be maintained for 24 hours and Step 3 of Procedure shall be omitted.

4.10.1.2. High. The equipment shall be tested in accordance with Method 501.2, Procedure I of MIL-STD-810. The storage temperature shall be +160°F, (+71.1°C) and the operating temperature shall be +125°F, (+51.6°C).

4.10.2. Humidity. The equipment shall be tested in accordance with Method 507.2, Procedure II of MIL-STD-810. Step 2 and 3 may be omitted provided that humidity testing commences immediately following the temperature test and the test hardware has been located (after temperature testing) in an environment (factory floor, test chamber, etc) with temperature and humidity conditions not exceeding those stipulated in Step 3. Twice, during each humidity cycle, the measurements specified in the test plan shall be made as follows: once during the 24th to 28th hour period and once during the 44th to 48th hour period. The equipment shall remain in the humidity chamber for measurements and shall not be powered except while measurements are being taken.

4.10.2.1. Equipment failures during/after humidity. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If any equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. At the conclusion of these tests, the equipment shall be examined and there shall be no visible evidence of corrosion, rust or deterioration of finishes or changes in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service.

4.10.3. Altitude. The equipment shall be subjected to the test of Method 500.2, Procedure I, of MIL-STD-810, including non-operational test at 12.19 km (40,000 feet) except the operational altitude shall be 3.05 km (10,000 feet). The equipment shall be held at 12.19 km (40,000 feet) level non-operating for one hour. The rate of change of altitude shall not exceed 2500 feet per minute. The chamber may be returned to ambient atmospheric pressure for frequency changes.

4.10.4. Bench handling. The equipment shall be subjected to the test of Method 516.3, of MIL-STD-810.

4.10.5. Bounce, vehicular. The equipment shall be tested in accordance with Method 514.3 of MIL-STD-810. There shall be no physical damage to the equipment.

4.10.6. Fungus test. The equipment (including accessories) in its normal operating position shall be tested in accordance with Method 508.3, Procedure I of MIL-STD-810. The equipment, unless hermetically sealed, shall be opened and all surfaces, parts, and materials, on and under the chassis and the interior of the cabinets shall be sprayed with the spore suspension. Demountable chassis, modules, and electronic circuit board assemblies shall be removed and opened as applicable for thorough coverage of the fungi spore. The equipment shall remain in an open condition during the test. Operation is not required. At the conclusion of the test, the equipment shall be examined and

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there shall be no visible evidence of deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service. (See 3.10.6).

4.11. Reliability demonstration test. The radio set shall be given a Reliability Demonstration Test in accordance with the accept/reject criteria of test plan XX of MIL-STD-781. Specified MTBF (θ^o) shall be equal to or greater than the 600 hour MTFB of the unmodified AN/GRC-144. The equipment to be tested are the transmitter and receiver (and the converter in the SRWBR configuration). The minimum number shall be four.

4.11.1. Reliability test conditions. The test shall be conducted under the environment of test level B of MIL-STD-781 with the following additions and modifications.

4.11.1.1. Vibration. The peak acceleration value shall be 2.2G + 10% at any nonreasonant frequency between 20 and 40 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least ten (10) minutes during each hour of equipment operating time.

4.11.1.2. Equipment on-off cycling. The operating cycle shall be 3.5 hours on, 0.5 hours off.

4.11.1.3. Test duration. The test duration shall be 2340 equipment operating hours.

4.11.2. Parameters to be measured. The equipment undergoing test will be automatically monitored 24 hours a day for, bit errors, output power and radio set summary alarm. Periodically, the radio sets shall have their operating frequencies changed and shall be tested for BITE indications, metering and alarms.

4.11.3. Failure definition. Failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure from performance as required by the equipment specification. A relevant failure shall include failures as defined in paragraph 5.5.1 of MIL-STD-781 and any failure or degradation of government furnished equipment and government loaned equipment caused by deficiencies in contractor furnished equipment. Analysis of failures shall be in accordance with paragraph 5.5.2 of MIL-STD-781. Determination of compliance shall be in accordance with paragraph 5.4.8 of MIL-STD-781.

4.12. Electromagnetic interference test. The AN/GRC-144(V)3 shall be tested to verify that the EMI characteristics requirements of MIL-STD-461 as specified in 3.11 herein are met. Test Methods and procedures shall be in accordance with MIL-STD-462.

4.13. Interchangeability. When a mechanical gage or an electrical test fixture is referenced in the contract, they shall be used to determine compliance with the mechanical or electrical interchangeability requirement, respectively. For those modules or sub-assemblies not having a gage or test fixture the following will apply:

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4.13.1. Mechanical. The dimensions listed below shall be gage or measured to determine conformance to the mechanical interchangeability requirement of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Locations of connectors, locking pins, fasteners and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.13.2. Electrical. Plug-in assemblies, including spares, shall meet the requirements of their respective test requirements specification drawings.

4.14. Operational test. The operational test shall be performed to determine that the radio set is operating satisfactorily before, during and after environmental tests as applicable, and prior to packaging. (See SM-A-956758).

5. PACKING

5.1. Packing requirements. The packing requirements for the desired level(s) of protection shall be as specified by the acquisition activity. (See 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The radio set has been designed in accordance with TT-B1-2205-0027. This specification shall be the governing document in the event any changes are required to the radio set configuration.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced. (See 2.1).

c. Level A, level B or commercial packaging. (See Section 5).

d. When rough handling test is required.

e. Marking and shipping of samples.

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f. Place of final inspection.

g. Technical literature required, test tools, and running spares.

h. First article inspection. (See 3.1).

i. Length of cables (if any are required).

6.3. First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4. Group C inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3).

6.5. Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example AN/GRC-144(V)3. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract.

6.6. Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum use of the contractor's quality history of the product.

6.7. Location of operational test. It is desirable that the operational test be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging. (See 4.14).

6.8. Subject term (keyword) listing.

Equipment, communications, LOS, HI-CAP
Equipment, communications, wide band, short range

6.9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX

RADIO SET AN/GRC-144(V)4

10. SCOPE

10.1. Scope. This appendix establishes the requirements for manufacture and acceptance of Radio Set AN/GRC-144(V)4 (SRWBR) which operates as a Short Range Wide Band Radio at data rates up to 18.72 Mb/s in the 14.4 - 15.0 GHz. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1. Government documents.

20.1.1. Specifications, standards, and handbooks. In addition to the documents specified in 2.1 the following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 60.2).

SPECIFICATIONS

MILITARY

MIL-A-3933 Attenuators, Fixed, General Specification for

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

20.1.2. Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Communications-Electronic Command

SM-D-956417	Filter, BP 70 MHz
SM-C-956419	Filter, LP 5.3 MHz
SM-C-956421	Filter, LP 13 MHz
SM-D-956431	Attenuator Assembly
SM-D-956462	Equalizer - Filter
SM-A-966926	SRWBR Antenna Applique Kit
DL-SM-B-956717	Applique Kit, 15 GHz Antenna
DL-SM-B-956774	Radio Set, SRWBR
SM-C-956780	Cable Assembly

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(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

30. REQUIREMENTS

30.1. First article. When specified (See 40.3) the contractor shall furnish sample units for first article inspection and approval. (See 40.3 and 6.2).

30.2. Construction. Radio set AN/GRC-144(V)4 shall be constructed in accordance with this specification and drawing and data list DL-SM-B-956774 and DL-SM-B-956717.

30.3. AN/GRC-144(V)3 conformance. The AN/GRC-144(V)4 shall meet the requirements of 3.3 through 3.15.

30.4. AN/GRC-144(V)4 operational at 14.4 to 15.0 GHz.

30.4.1. Frequency accuracy. When driven by the transmitter the output frequency shall be within ± 120 kHz of the assigned frequency.

30.4.2. Power output capability. The converter power output measured at the 15 GHz converter waveguide flange shall be equal to or greater than 250 mw through the frequency range 14.4 to 15.0 GHz for transmitter drive input levels equal to or greater than 200 mw in the frequency range 4.8 to 5.0 GHz.

30.4.3. Power output stability. With constant transmitter drive input, the converter output power shall be within +1dB of its initial value during any 24 hour period.

30.4.4. Receiver frequency range. In the receive direction with an input in the 14,400 to 14,9799 GHz range, the converter output frequency shall be in the 4.400 to 4.9799 GHz range, with a local oscillator (LO) of 10.000 GHz. In the 14.980 to 15.000 GHz range, the output frequency shall be 4.960 to 4.980 GHz with an LO of 10.020 GHz.

30.4.5. Frequency stability. The converter receive section local oscillator frequency stability shall be within ± 0.001 percent.

30.4.6. Output level and stability. In the receive direction, over the frequency band of 14.4 to 15.0 GHz the minimum converter gain shall be 5 dB and the converter output stability shall be ± 1 dB during a 24 hour period at room ambient.

30.4.7. Noise figure. The converter noise figure of the AN/GRC-144(V)4 radio set, measured at the 15 GHz converter waveguide flange, shall not exceed 10 dB. 30.4.8. IF bandwidth. The 3 dB, 70 MHz receive IF bandwidth as measured from the AN/GRC-144(V)4, 5 GHz receiver input to the IF amplifier output shall be as follows:

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Traffic rate (Mb/s)	Bandwidth (MHz)
9.360	10.6 ± 2
18.720	20.0 ± 4

30.4.9. Carrier level threshold. The threshold for the receiver with converter, measured at the converter flange shall be reached by the carrier levels listed for the following rates:

Traffic rate (Mb/s)	Level (dBm)
4.096, 4.608, 4.9152	-84.0
9.360	-81.0
18.720	-78.0

30.4.10. Signal-to-noise ratio. The receiver output of the AN/GRC-144(V)4 radio set shall have a minimum of 20 dB signal-to-noise (peak-to-peak signal to RMS noise) ratio performance for digital operation, with the following simulated path losses between converters:

Traffic rate (Mb/s)	Path loss (dB)
9.360	101.8
18.720	96.5

30.4.11. Pulse response. Eye pattern level comparison between 4.9152 Mb/s (96 CH) and 18.72 Mb/s data rate operations shall not exceed a 20% difference when the respective pseudo-random pulse patterns are transmitted at rates deviation and at the same carrier-to-noise ratio.

30.4.12. System bandwidth. The overall transmitter through receiver bandwidth shall be optimized in two bandwidths (see table AI) for transmission of data at 9.36 and 18.72 Mb/s.

Traffic rate (Mb/s)	Bandwidth (MHz)
9.360	3.8 MHz, 3 dB ± 0.6 dB
18.720	7.0 MHz, 3 dB ± 1.0 dB

TABLE XIII. Mode module implementation.

Transmitter Unit 1		
Radio Usage	Ref Dis A3	Ref Dis A6
9.36 Mb/s	SM-D-956431 Attenuator Assy	SM-C-956780 Cable Assy
18.72 Mb/s	SM-D-956431 Attenuator Assy	SM-C-956780 Cable Assy

Receiver Unit 2			
Radio Usage	Ref Dis A3	Ref Dis A6	
9.36 Mb/s	SM-C-956419 Fltr, LP, 5.3 MHz	SM-D-956417 Fltr, BP, 70 MHz (B=10.6 MHz)	M3933/18-01 Fixed Atten

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TABLE XIII. Mode module implementation - Continued.

Radio Usage	Receiver Unit 2 Ref Dis A3	Ref Dis A6	
18.72 Mb/s	SM-C-956421 Flt, LP, 13 MHz	SM-D-956462 Equalizer Filter	M3933/18-01 Fixed Atten
*Connects between 2 FL5 and 2A7.			

30.4.13. Output level and stability. The baseband output level of the receiver with converter operating into a 50 ohm termination shall be 1.0 volt peak-to-peak minimum. The output level shall not vary by more than ± 1 dB with input levels from -38.5 dBm to -73.5 kBm at 18.72 Mb/s and -38.5 to -77.8 dBm at 9.36 and 4.9152 Mb/s.

30.4.14. Alarm systems. The converter shall properly operate a visual go-no-go summary alarm which monitors loss of RF output power, excess reflected power, loss of local oscillator power, loss of air flow and over temperature. The summary alarm shall be integrated with the AN/GRC-144 summary alarm to provide audible indication of multiplexer-converter failure, as well as a contact closure for interface with the Orderwire Control Unit.

30.4.15. Metering. Metering shall properly operate via tip jacks for use in conjunction with the R-1467(P)A/GRC-144(V)4 receiver front panel meter and test lead for the following levels: Forward and reflected power, local oscillator power, ± 100 VDC and ± 28 VDC.

30.4.16. Bit error rate. The transmitter, receiver and 15 GHz converter operating in a closed loop through the digital data combiner shall have bit errors of less than 1 in 10^{5} for the following data rates and RF simulated path losses in the 14.4 to 15.0 GHz band.

Data rates (Mb/s)	Simulated path loss (db)
9.36 Mb/s	101.8 + 0, -1
18.72 Mb/s	96.5 + 0, -1

30.5. Antenna Assembly AS-3398/GRC-144(V).

30.5.1. Gain. The gain of the antenna relative to an isotropic radiator shall be at least 43 dB over the range 14.4 to 15.0 GHz.

30.5.2. Minor lobes. The ratio of the major lobe to any minor lobe shall be at least 20 dB at any frequency between 14.4 to 15.0 GHz.

30.5.3. VSWR. The VSWR shall not exceed 1.15:1 over the frequency range 14.4 to 15.0 GHz, referred to the feedhorn input when the feedhorn is mounted in the parabolic refector under free space conditions.

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30.5.4. Feedhorn de-icer. The heater circuit shall operate on 115 Vac, 60 cycles and shall draw between 141 and 173 milliampere.

30.5.5. Waveguide. The insertion loss of each 90 foot run shall be 5.7 dB maximum.

30.5.5.1. Waveguide pressurization. The waveguide run shall be capable of pressurization up to 5 pounds per square inch absolute (psia), with a pressure relief valve mounted in the transition behind the feedhorn and a schrader valve near the shelter waveguide entry panel inside the shelter (not part of the radio set).

30.6. 15 GHz system. The following requirements are applicable for both narrowband and wideband operation at 15 GHz:

a. 3.6.1.1. Analog Voice Orderwire (AVOW) signal-to-noise ratio.

b. 3.6.1.2 Analog orderwire levels.

c. 3.6.1.4. Analog orderwire peak limiting.

d. 3.6.1.5. Analog orderwire distortion.

e. 3.6.1.3. Analog orderwire frequency response.

f. 3.6.2. Output capability and stability.

30.7. Operational test (Radio Set). The operational test of 4.14 shall be performed to determine that the radio set is operating satisfactorily before, during and after environmental tests and to determine that each radio set is operating satisfactorily prior to packaging.

30.8. Marking. In addition to the requirements of 3.5, the Converter Multiplier CV-3633/GRC-144(V) shall be serial numbered.

30.9. Electromagnetic interference (EMI) characteristics. The equipment shall meet the following EMI and susceptibility requirements of MIL-STD-461.

Test Method	Description	Frequency Range, Hz	Notes
CE01	Primary Powerline Emissions	30 to 20k	a, b, c
CE02	Control and Signal Line Emissions	30 to 20k	
CE03	Primary Powerline Emissions	20k to 50M	
CE04	Control and Signal Line Emissions	20K to 50M	
CE06	Spurs at the Antenna Terminal	3.5G-18G	d
CS01	Low Frequency Susceptibility, Primary Power Lines	30 to 50k	
CS02	High Frequency Susceptibility, Primary Power Lines	50 to 400M	
CS03	Intermodulation Susceptibility in RF Receivers	3.5G-18G	

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CS04	Input Selectivity Susceptibility in RF Receivers	3.5G-18G	
CS05	Cross-Modulation Susceptibility in RF Receivers	3.5G-18G	
CS06	Spike Susceptibility in Primary Power Lines	10μ, + 110V	
RE02	E-Field Emissions from Case and Cabling	14k to 18G	e
RS03	Case and Cable Susceptibility to External E-Fields	14k to 18G	

NOTES:

a. CE01 Broadband emission limit for RCVR Cabinet: +97 dBua/20 kHz, based on a 2.3-ampere primary current.

b. CE01 Broadband emission limit for XMTR Cabinet: +98 dBua/20 kHz, base on a 3.0-amperes primary current.

c. Additional relief, CE01 Broadband emission limit: Additional relief shall be afforded the limits of (1) and (2) above, in accordance with the following formula:

$$\text{Delta relief, dB} = 20 \log_{10} \frac{\text{Primary current of SRWBR Cabinet}}{\text{Applicable current of (1)or(2),above}}$$

d. Test Method CE06: Reference MIL-STD-461. Paragraphs 6.3.1 and 6.3.2 are not applicable to this test. Paragraph 6.3.3 is applicable to this test.

e. Test Method RE02, Narrowband: The following relief factors are in effect for this test method:

(1)	14 kHz to 1 GHz:	6 dB
(2)	1 GHz to 5 GHz:	20 dB for all emissions except the C-Band transmit signal, and the 4th harmonic of the Transmitter's frequency synthesizer.
(3)	C-Band Transmit Signal:	30 dB
(4)	4th Harmonic of the Transmitter's Frequency Synthesizer:	25 dB

40.0 QUALITY ASSURANCE PROVISIONS

40.1. Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the contractor is responsible for performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

40.1.1. Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

40.2. Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 40.3).

b. Inspection covered by subsidiary documents. (See 40.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 40.5).

40.3. First article inspection. This inspection shall consist of all the inspections specified in Table XIV and shall be performed in the order shown on the number of sample units specified in the contract.

TABLE XIV. Order of first article inspection.

INSPECTION	REQ. PARA	INSP. PARA	UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5-8	Notes
1. Inspection covered by subsidiary documents.		40.4	Inspections to be performed on all units.					c
2. Bounce Pre-conditioning	3.9.1	4.7.1	Inspections to be performed on all units.					c
3. Electrical Preconditioning	3.9.2	4.7.2	Inspections to be performed on all units.					c
4. Group A Inspection	See Table XV		Inspections to be performed on all units.					c

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TABLE XIV. Order of first article inspection - Continued.

INSPECTION	REQ. PARA	INSP. PARA	UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5-8	Notes
5.Group B Inspection	See Table XVI		1	1				a
6. Group C Inspection (Electrical)	See Table XVII		1	1				a
7. EMI	30.9	4.12	2					-
8. Service Conditions								
Non-damaging								
Low Temp.	3.11.1.1	4.10.1.1		2				-
High Temp.	3.11.1.2	4.10.1.2		3				-
Altitude	3.11.3	4.10.3			1			d
Bench Handling	3.11.4	4.10.4			2			c
Potentially damaging:								
Humidity	3.11.2	4.10.2		4				c
Bounce Veh	3.10.5	4.10.5			3			d
Fungus	3.11.6	4.10.6				1		d
9. Reliability	3.10	4.11					1	b

NOTES:

a. The inspections 1 through 4, in the order shown, shall be performed on all the first article units before subjecting them to any other inspection requirements specified in the table except that Group B and Group C inspections may be commingled with Group A inspections to reduce the number of test set-ups. Unless otherwise noted the unit is an AN/GRC-144 modified or HI-CAP LOS/SRWBR operation or a new AN/GRC-144(V)4.

b. Reliability qualification (demonstration) testing for the AN/GRC-144(V)4 shall be conducted on first article units (4 units shall be the sample size required for qualification testing) in accordance with 4.11.

c. New radio set or new and modified modules only.

d. New radio set only, no test required for modified AN/GRC-144.

40.4. Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before packaging:

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Item	Where required
Finish	3.4
Marking	30.8
Safety	3.13
Workmanship	3.15

40.5. Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 40.4 and 40.5.1 through 40.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure.

40.5.1. Group A inspection. Each radio set on contract shall be inspected for conformance to the inspections specified in Table XV. The order of inspection of the subassemblies, assemblies and units shall be as follows:

a. Prior to their assembly into the end equipment, the radio set subassemblies on order (new and modified subassemblies only) shall be subjected to a 100% visual and mechanical inspection by the contractor for conformance with the inspection requirements provided in Table XV.

b. Assemblies which have passed the above inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 3.9 and a 100% visual/mechanical and electrical inspection in accordance with Table XV.

c. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection in accordance with Tables XV-1 and XV-2. Group A electrical inspection shall be performed in any order which is satisfactory to the Government, except that the operational test 4.14 shall be performed last (See 6.7).

NOTE: Assemblies are not required to undergo further visual/mechanical inspection as part of the contractor's 100% Group A inspection of the end equipment.

TABLE XV. Group A inspection.

Inspection	Required Paragraph	Test Procedures
Visual/mechanical:		
Cabinets, elec equipment	3.13 & 3.15	4.6
Printed wiring assemblies	3.13 & 3.15	4.6
Electrical:		
AN/GRC-144(V)3 conformance	SEE TABLE IX	
AN/GRC-144(V)4 operation at		

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TABLE XV. Group A inspection - Continued.

Inspection	Required Paragraph	Test Procedures
14.4 to 15.0 GHz: Power output capability	30.4.2	SM-A-956748
Receive frequency range	30.4.4	SM-A-956748
Noise figure	30.4.7	SM-A-956748
Carrier level threshold	30.4.9	SM-A-956748
Alarm systems	30.4.14	SM-A-956748
Metering	30.4.15	SM-A-956748
Bit error rate	30.4.16	SM-A-956742
Analog Voice Orderwire (AVOW) signal-to-noise ratio	30.6	SM-A-956742

Note a: Defects in all characteristics other than visual/mechanical are considered major.

TABLE XV-1. Sampling plan for visual and mechanical inspections.

Lot Size	Sample Size
2 to 8	*
9 to 15	8
16 to 25	8
26 to 50	8
51 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27
1201 to 3200	35
3201 to 10,000	38
10,001 to 35,000	46
35,001 to 150,000	56
150,001 and over	64

NOTES:
a. * Indicates entire lot must be inspected
b. Acceptance number in all cases is zero.

TABLE XV-2. Sampling plan for electrical inspections.

Lot Size	Sample Size
2 to 8	5
9 to 15	5
16 to 25	5
26 to 50	5
51 to 90	7
91 to 150	11

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TABLE XV-2. Sampling plan for electrical inspections - Continued.

Lot Size	Sample Size
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40

NOTE:
a. Acceptance number in all cases is zero.

40.5.2. Group B inspection. Group B inspection shall be performed on lots that have passed Group A inspection. This inspection shall conform to Table XVI and shall be performed in any order satisfactory to the Government. Each lot shall be subjected to sampling inspection in accordance with Table XVI-1.

TABLE XVI. Group B inspection.

Inspection	Required Paragraph	Test Procedure
Interchangeability	3.14	4.13
IF bandwidth	30.4.8	SM-A-956750
Analog orderwire levels	30.6	SM-A-956744

TABLE XVI-1. Sampling plan for group B inspections.

Lot Size	Sample Size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29

NOTE:
a. Acceptance number in all cases is zero.

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40.5.3. Group C inspection. Group C inspection shall be performed on units that have passed Group A and B inspection. The inspections shall consist of the inspections specified in Table XVI. Samples shall be selected in accordance with 40.5.3.1.

40.5.3.1. Sampling for Group C inspection.

40.5.3.1.1. Subgroup 1. For this subgroup, two (2) units from the first month's production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two (2) units from each successive 100, or two (2) units per month if less than 100 are produced month, shall be selected. For this subgroup the maximum number of units to be selected and tested in a single month shall not exceed eight (8) units.

40.5.3.1.2. Subgroup 2. For this subgroup, two (2) units from the first month's production shall be selected for the first Group C inspections. For subsequent Group C inspections, two (2) units shall be selected and tested during each two (2) month period throughout production.

40.5.3.1.3. Subgroup 3. For this subgroup, excluding first article testing, one (1) unit shall be selected at random and tested at the midpoint of the production schedule.

TABLE XVII. Group C inspection.

Inspection	Required Paragraph	Test Procedure
Subgroup I		
Frequency accuracy	30.4.1	SM-A-956746
Power output stability	30.4.3	SM-A-956746
Frequency stability	30.4.5	SM-A-956752
Output level & stability	30.4.6	SM-A-956752
Signal-to-noise ratio	30.4.10	SM-A-956752
Pulse response	30.4.11	SM-A-956752
System bandwidth	30.4.12	SM-A-956752
Output level & stability	30.4.13	SM-A-956752
VSWR	30.5.3	SM-A-966926
Feedhorn de-icer	30.5.4	SM-A-966926
Waveguide	30.5.5	SM-A-966926
Waveguide pressurization	30.5.5.1	SM-A-966926
Analog orderwire peak limiting	30.6	SM-A-956746
Analog orderwire distortion	30.6	SM-A-956746
Analog orderwire frequency response	30.6	SM-A-956746
Output capability and stability (Baseband)	30.6	SM-A-956752
Video output impedance	3.6.1.6	SM-A-956746
Analog Orderwire (AVOW) S/N ratio with supply line variations	3.6.1.7	SM-A-956746
Digital data characteristics:	3.6.2	SM-A-956758
Data rates	3.6.2.1	SM-A-956758
Format	3.6.2.2	SM-A-956758
Driver output	3.6.2.3	SM-A-956758

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TABLE XVII. Group C inspection - Continued.

Inspection	Required Paragraph	Test Procedure
Rise and fall times	3.6.2.4	SM-A-956758
Clock data phasing	3.6.2.5	SM-A-956758
Driver offset	3.6.2.6	SM-A-956758
Signal balance	3.6.2.7	SM-A-956758
Ringing	3.6.2.8	SM-A-956758
Receiver sensitivity	3.6.2.9	SM-A-956758
Input load impedance	3.6.2.10	SM-A-956758
Digital Voice Orderwire (DVOW)	3.6.2.11	SM-A-956758
Rate	3.6.2.11.1	SM-A-956758
Format	3.6.2.11.2	SM-A-956758
Data Orderwire	3.6.2.12	SM-A-956758
Rate	3.6.2.12.1	SM-A-956758
Format	3.6.2.12.2	SM-A-956758
Power output with supply line variations	3.6.6.1	
Radio test set operation	3.6.8	SM-A-956746
Carrier Level threshold with supply line variations	3.7.6.1	See Note a
Frequency accuracy	3.7.7	SM-A-956752
Frequency accuracy with supply line variations	3.7.7.1	See Note b
Transmitter and receiver system PCM response	3.8.5	SM-A-956752
Video frequency response	3.8.6	SM-A-956752
DDC operation with supply line variations	3.8.8	SM-A-956752
Subgroup 2 (See Notes c & d)		
Temperature	3.11.1	
High	3.11.1.2	4.10.1.2
Low	3.11.1.1	4.10.1.1
Humidity	3.11.2	4.10.2
Bench handling	3.11.4	4.10.4
Bounce vehicular	3.11.5	4.10.5
Electromagnetic interference	3.12	4.12
Subgroup 3		
Altitude	3.11.3	4.10.3
Fungus	3.11.6	4.10.6
Electromagnetic interference	3.12	4.12
Gain	30.5.1	SM-A-966926
Minor lobes	30.5.2	SM-A-966926

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APPENDIX

Notes: a. Test to be performed in accordance with SM-A-956748 except that the ac power source shall be adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz.

b. Test to be performed in accordance with SM-A-956752 except that the ac power source shall be adjusted to 110 Vac, 47 Hz and then to 120 Vac, 63 Hz.

c. Both units shall be inspected for the following EMI characteristics:

Receive mode: CE06, RE02, and RS03 Transmit mode: RE02

d. Perform the operational test before, during (as applicable) and after environmental test in accordance with the applicable test procedure.

After EMI testing one unit shall be subjected to the High temperature, Low temperature Humidity tests in that order. The remaining unit shall be subjected to the Bench handling and Bounce vehicular tests in that order.

40.5.3.2. Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

40.5.4. Reinspection of conforming Groups B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

40.5.5. Group D inspection. This inspection shall consist of the tests specified in Table XVIII and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE XVIII. Group D inspection.

Inspection	Required Paragraph	Test Procedure
Reliability	3.10	SM-A-966924

40.5.5.1. Sampling for Group D inspection. Four samples shall be selected at random from the monthly production lot.

40.5.5.2. Group D failures. Actions required relative to Group D failures shall be as specified in the contract.

40.5.5.3. Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on contract provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

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APPENDIX

50. PACKING

50.1. Packing requirements. The packing requirements for the desired level(s) of protection shall be as specified by the acquisition activity. (See 6.2).

60. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

60.1. AN/GRC-144(V)4 design. The radio set has been designed in accordance with TT-B1-2205-0027. This specification shall be the governing document in the event any changes are required to the radio set configuration.

60.2. Acquisition requirements. (See 6.2).

60.3. First article. (See 40.3 and 6.3).

60.4. Environmental. (See 6.4).

60.5. Group C inspection. (See 40.5.3 and 6.5).

60.6. Nomenclature. (See 6.6).

60.7. Verification. (See 6.7).

60.8. Location of operational test. (See 6.8).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-49226a(CR)	90/07/31

3. DOCUMENT TITLE: RADIO SET AN/GRC-144( )(V)3 and (V)4

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY CECOM	(1) Commercial (908)532-5851	(2) AUTOVON 992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TO FORT MONMOUTH, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 . AUTOVON 289-2340	

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